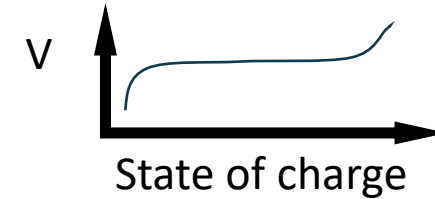
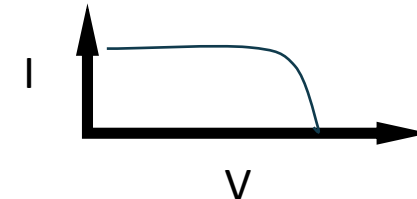




- 3 ports : 2 AC – 1DC $\rightarrow$ AD/DC
- Power can flow in both direction with respect to DC port $\rightarrow$ Bidirectional converter or Inverter and rectifier (several names valid)
- Probably* at least few IGBT or MOSFET in the converter circuit $\rightarrow$ Force Commutated Converter
- A battery can be model as a constant voltage source, and as P flows in both direction current will have + and - polarity $\rightarrow$ VSC



- 2 ports : 1 AC – 1DC $\rightarrow$ AD/DC
- Power can flow in both direction with respect to DC port **but lighting the sky with bad LED is a very useless usage ;)** , so... $\rightarrow$ Inverter
- Probably* IGBT or MOSFET based inverter circuit $\rightarrow$ Force Commutated Converter
- PV panels ?? $\rightarrow$ Not so obvious for now ...



*Manufacturers do not publish the schematic of their converter topology, and reverse engineering is quite complicated